Lesson 11:

Graphs and Trending



IGSS

Interactive Graphical
SCADA System

YouTube
Broadcast Yourself™

Contents of this lesson



Topics:

- The window-type graph
- The embedded graph
- Create dynamic graphs

The window-type graph

- A chart showing up to ten IGSS objects
- Analog, digital, table and counter objects
- Data source may be LOG data or BCL data (reduced values)



Graph subtypes



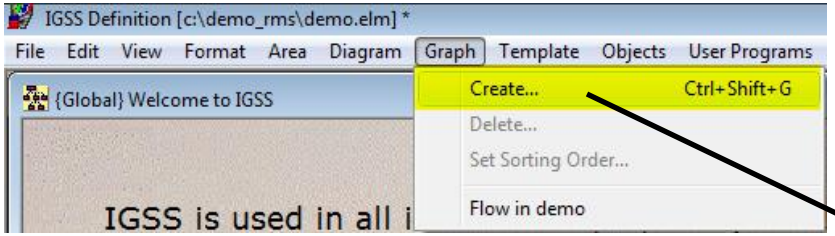
1. A predefined graph is defined by the system designer in Definition.

2. A dynamic graph is created on-the-fly by the operator during supervision. A dynamic graph can be saved and re-opened at a later time.

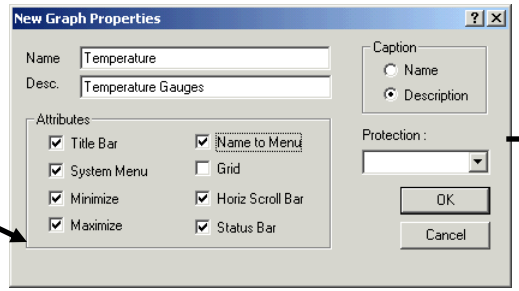
3. A static graph is a snapshot of a graph saved as a file.

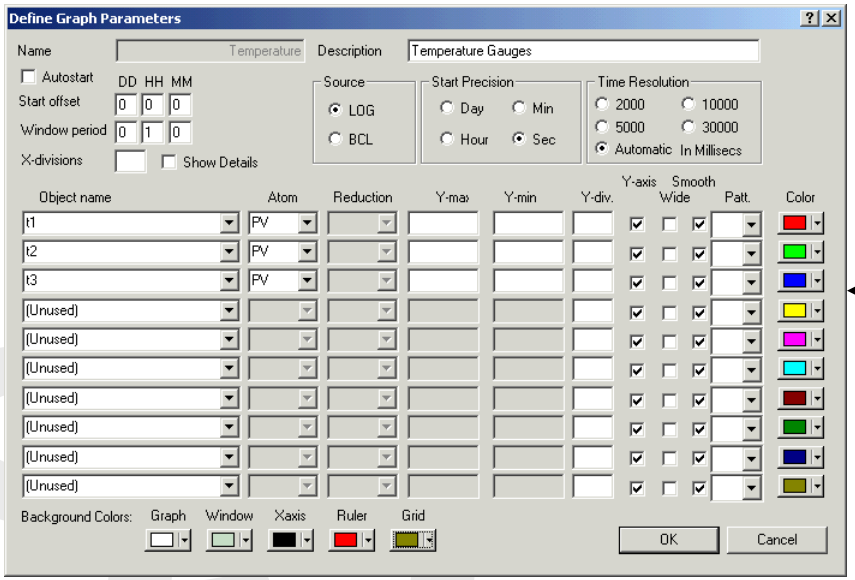
Create a predefined graph

STEP 1: Select ”Create” in Graph menu.



STEP 2:
Choose window elements and ”Name to menu”

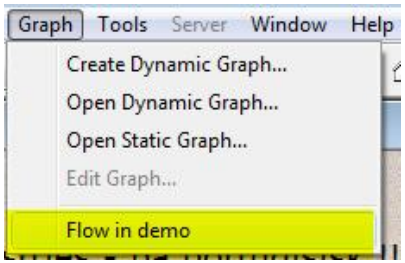




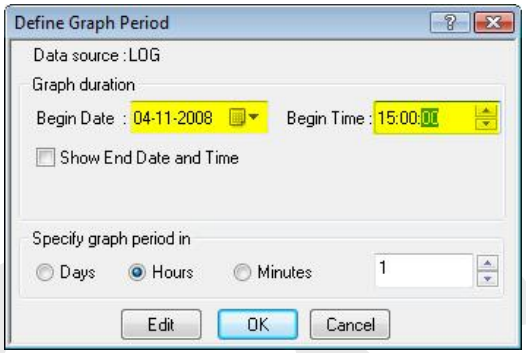
STEP 3:
Choose objects to show, colors and other parameters.

How the user launches the graph

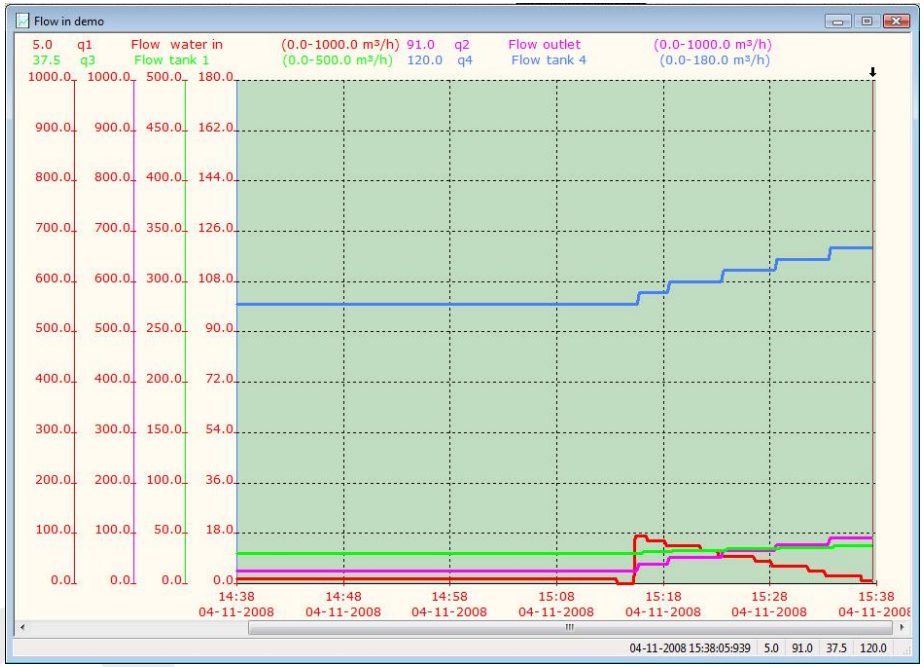
1. Activate the graph.



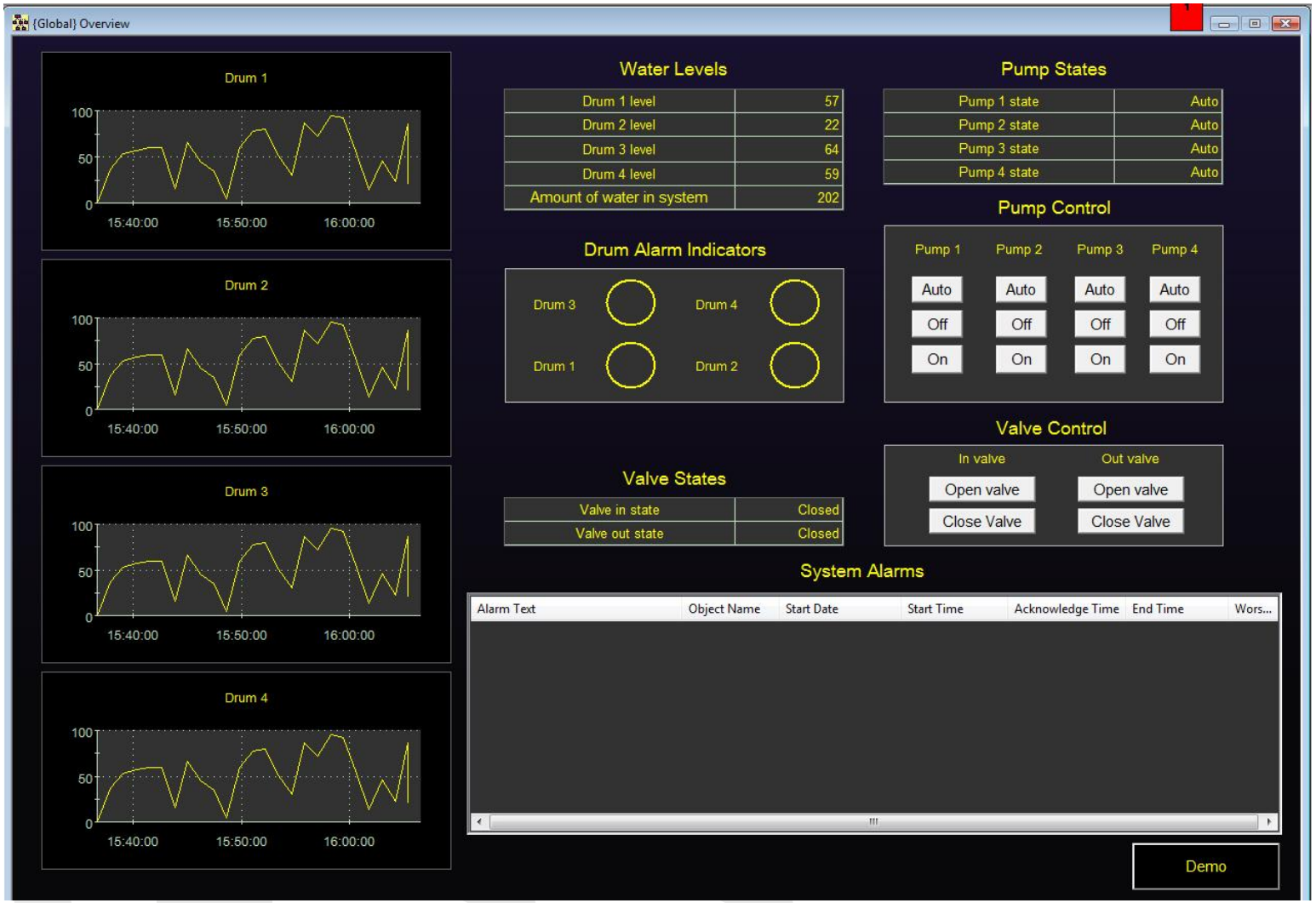
2. Choose the graph period.



3. The graph appears.

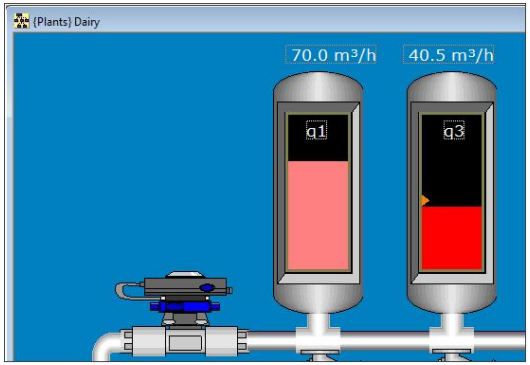


The embedded graph

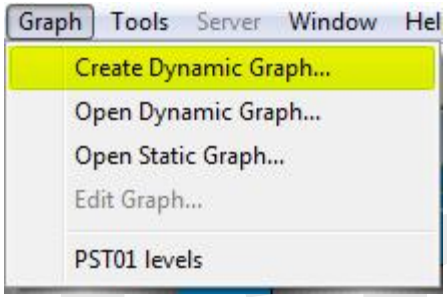


Create a dynamic graph

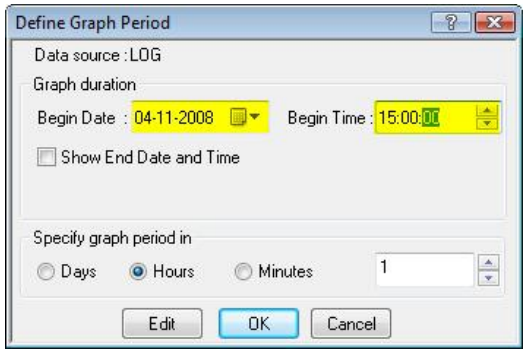
1. Select objects on diagram.



2. Select ”Create Dynamic Graph”.



3. Choose period.

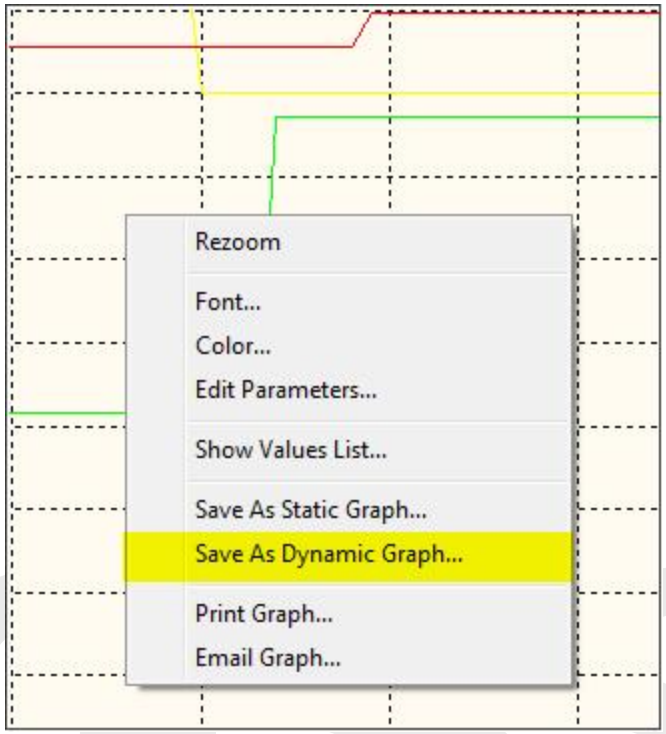


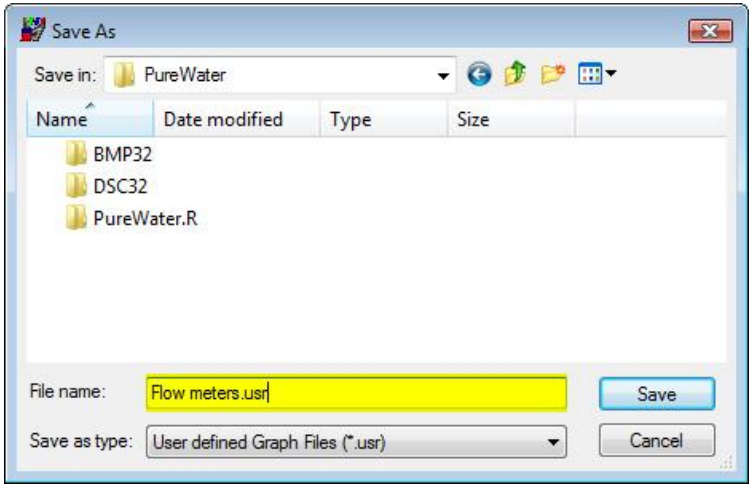
4. The graph appears.



Save a dynamic graph

1. Right-click the graph.
2. Select ”Save as Dynamic Graph”.
3. Give the graph a name.
4. All operators can now open the dynamic graph.





Try it yourself ...



Exercise: Create an embedded graph

Find presentations and exercises on www.7t.dk/igss



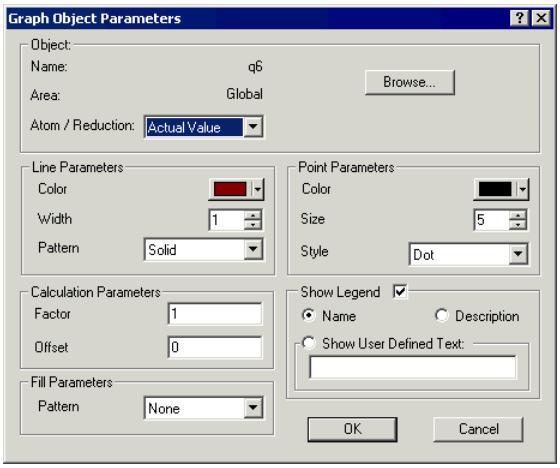
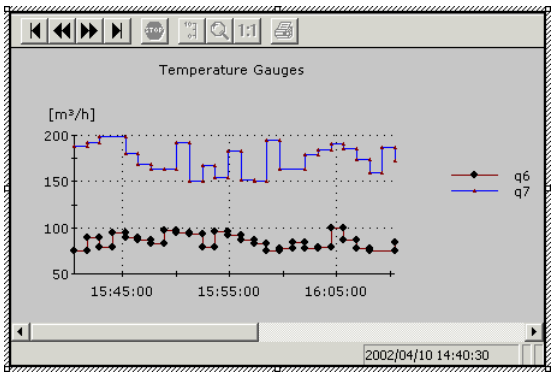
Exercise: Create an Embedded Graph

Purpose This exercise demonstrates how you can use the new graph type to create traditional time plots.

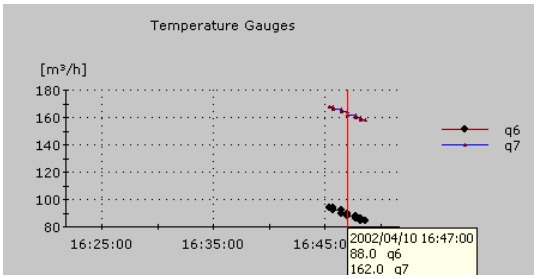
Duration Max. 20 min.

Task 1: Create a time plot In this first task we will embed a graph showing the process values from the two flow gauges, **q1** and **q2**.

Step	Action
1	Create a new diagram and name it `Graphs` and choose a color for the background. Remember to enable the **Name to menu** option.
2	If the **Drawing** toolbar is not shown already, choose **Drawing toolbar** in the **View** menu.
3	Click the new graph icon in the **Drawing** toolbar and point at the middle of the **Graph** diagram. When you click on the diagram, an empty graph is inserted.
4	Double-click inside the empty graph to bring up the **Graph Properties** dialogue.
5	Click the **Add** button to add the first object to the graph.
6	In the **Graph Object Parameters** dialogue, click the **Browse** button.
7	Find and select the object **q1** and click **Open / Select**.

8	Do the following: • Under **Line Parameters** select a brown color • Under **Point Parameters** change the **Size** to "5" and the **Style** to **Dot**. • Select the **Show Legend** check box and click **OK**. 
9	Repeat steps 4 – 7 for the object **q2** and choose a new line color and point style.
10	Click the **General** tab. Under **Graph Title** type "**Flow Gauges**" and click **OK**. The new graph appears showing random values for the included objects. 
11	Install the configuration.
12	Start the configuration.
13	Go to the **Graphs** diagram and check that the graph looks as expected.



14	Place the red marker above the incoming process values and notice that you can see the values in a small tooltip. 

YOU HAVE SUCCESSFULLY COMPLETED THE EXERCISE !!